Marcos García Blanco

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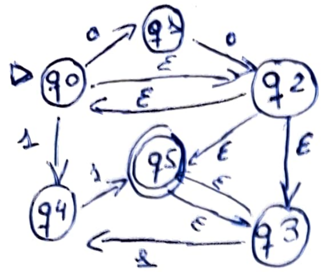


q_0^D = clausura-e(q_0) = $\{q_0, q_2, q_3, q_5\}$

mueve(q_0^D, 0) = $\{q_1\}$ clausura-e(q_1) = $\{q_1\} \equiv q_1^D$
mueve(q_0^D, 1) = $\{q_4\}$ clausura-e(q_4) = $\{q_4\} \equiv q_2^D$

mueve(q_1^D, 0) = $\{q_2\}$ clausura-e(q_2) = $\{q_2, q_0, q_3, q_5\} = q_0^D$
mueve(q_1^D, 1) = $\emptyset$

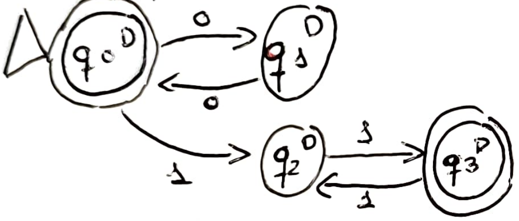
q_0 = A
q_1 = B
q_2 = C
q_3 = D
q_4 = E
q_5 = F*

AFD = $\{Q^D, E, \delta^D, q_0^D, F^D\}$

mueve(q_2^D, 0) = $\emptyset$
mueve(q_2^D, 1) = $\{q_5\}$ clausura-e(q_5) = $\{q_3, q_5\} \equiv q_3^D$

mueve(q_3^D, 0) = $\emptyset$
mueve(q_3^D, 1) = $\{q_4\}$ clausura-e(q_4) = q_2^D

AFD equivalente:



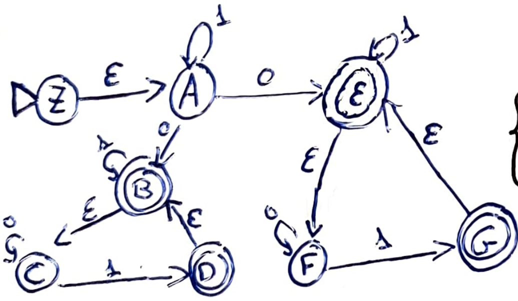
$Q^D = \{q_0^D, q_1^D, q_2^D, q_3^D\}$

$F^D = \{q_0^D, q_3^D\}$

Minimizar AFD:

$E_0 = \{\{q_0^D, q_3^D\}, \{q_1^D, q_2^D\}\}$
$E_1 = \{\{q_0^D\}, \{q_3^D\}, \{q_1^D\}, \{q_2^D\}\}$ ⇒ no se puede simplificar más

2.



Z^D = clausura-e(Z) = $\{Z, A\}$

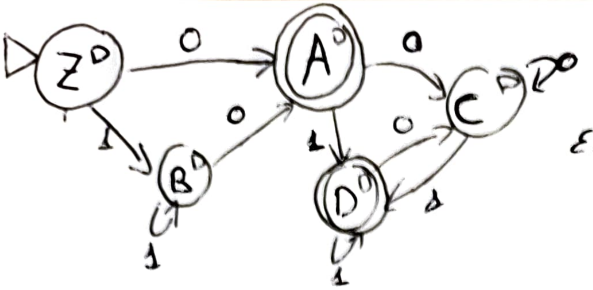
mueve(Z^D, 0) = $\{B, E\}$ clausura-e(B, E) = $\{B, E, C, F\} = A^D$
mueve(Z^D, 1) = $\{A\}$ clausura-e(A) = $\{A\}$ = B^D

mueve(A^D, 0) = $\{C, F\}$ clausura-e(C, F) = $\{C, F\} = C^D$
mueve(A^D, 1) = $\{B, E, D, G\}$
clausura-e(B, E, D, G) = $\{B, C, E, F, D, G\} = D^D$

mueve(B^D, 0) = $\{B, E\}$ clausura-e(B, E) = A^D
mueve(B^D, 1) = $\{A\}$ clausura-e(A) = B^D

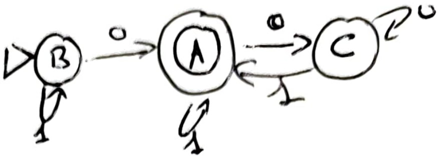
mueve(C^D, 0) = $\{C, F\}$ clausura-e(C, F) = C^D
mueve(C^D, 1) = $\{D, G\}$ clausura-e(D, G) = $\{D, G, B, E, F, C\} = D^D$

mueve(D^D, 0) = $\{C, F\}$ clausura-e(C, F) = C^D
mueve(D^D, 1) = $\{B, D, E, G\}$ clausura-e(B, D, E, G) = D^D

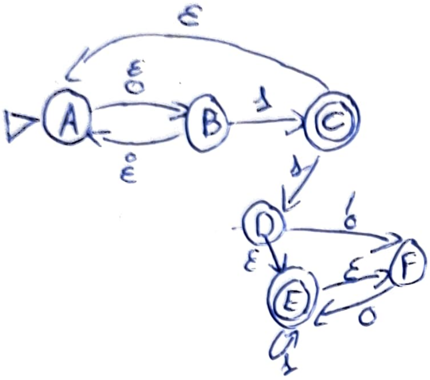


$\varepsilon_0 = \{\{A^D, D^D\}, \{Z^D, B^D, C^D\}\}$

$\varepsilon_1 = \{\underset{A}{\{A^D, D^D\}}, \underset{B}{\{Z^D, B^D\}}, \underset{C}{\{C^D\}}\}$



3.



A^D = clausura-e(A) = {A, B}

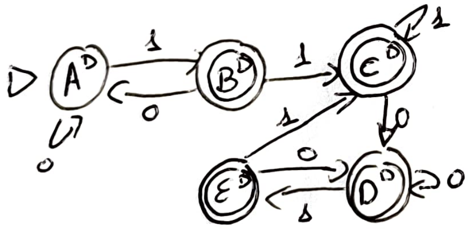
{ mueve (A^D, 0) = {B, A} clausura-e (A, B) = {A, B} = A^D
{ mueve (A^D, 1) = {C} clausura-e (C) = {A, B, C} = B^D

{ mueve (B^D, 0) = {A, B} clausura-e (A, B) = A^D
{ mueve (B^D, 1) = {C, D} clausura-e (C, D) = {A, B, C, D, E, F} = C^D

{ mueve (C^D, 0) = {A, B, F, E} clausura-e (A, B, F, E) = {A, B, F, E} = D^D
{ mueve (C^D, 1) = {C, D, F, E} clausura-e (C, D, F, E) = {C, D, F, E, A, B} = C^D

{ mueve (D^D, 0) = {B, A, E} clausura-e (B, A, E) = {B, A, E, F} = D^D
{ mueve (D^D, 1) = {C, E} clausura-e (C, E) = {C, E, A, B, F} = E^D

{ mueve (E^D, 0) = {B, A, E} clausura-e (B, A, E) = D^D
{ mueve (E^D, 1) = {E, C, D} clausura-e (E, C, D) = {E, C, D, A, B, F} = C^D



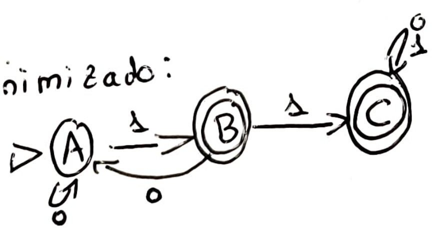
$\varepsilon_0 = \{\{B^D, C^D, D^D, E^D\}, \{A^D\}\}$

$\varepsilon_1 = \{\underset{B}{\{B^D\}}, \underset{C}{\{C^D, D^D, E^D\}}, \underset{A}{\{A^D\}}\}$

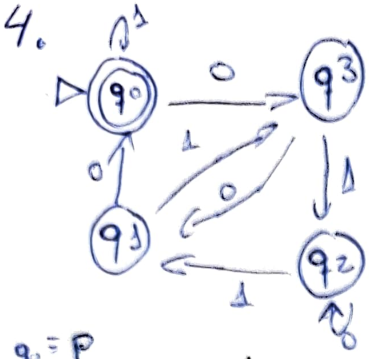
$C^D \xrightarrow{0} D^D$
$\xrightarrow{1} C^D$
$D^D \xrightarrow{0} D^D$
$\xrightarrow{1} E^D$
$E^D \xrightarrow{0} D^D$
$\xrightarrow{1} E^D$

Mismo grupo de ε_1

Minimizado:



4.



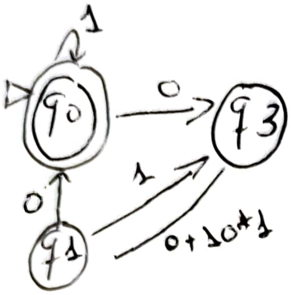
$E_R \equiv E_{R0}$

Eliminamos q2:

→q2	q2→
q3	q1

$\overline{q_3 q_1} = 10^*1$

$q_0 = P$
$q_1 = Q$
$q_2 = R$
$q_3 = S$

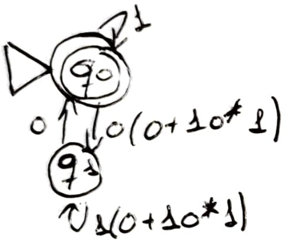


Eliminamos q3:

→q3	q3→
q0	q1
q1	

$\overline{q_0 q_1} = 0(0+10^*1)$

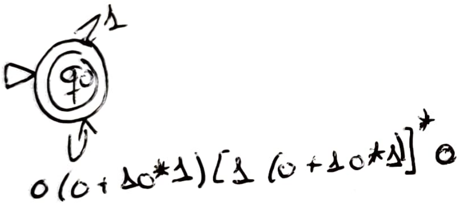
$\overline{q_3 q_1} = 1(0+10^*1)$



Eliminar q1:

→q1	q1→
q0	q0

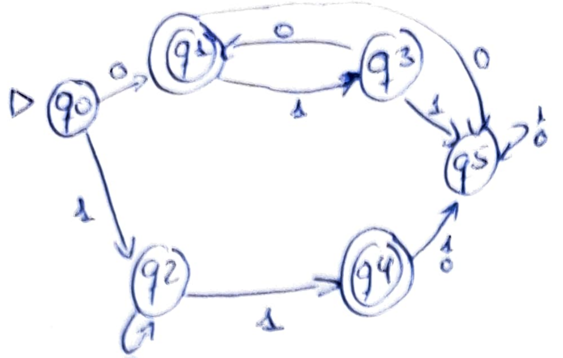
$\overline{q_0 q_0} \equiv 0(0+10^*1)[1(0+10^*1)]^*0$



~~$E_R \equiv E_{R0} = (1 + 0(0+10^*1)[1(0+10^*1)]^*0)^*$~~

$E_R \equiv E_{R0} = (1 + 0(0+10^*1)[1(0+10^*1)]^*0)^*$

5.



$ER \equiv ER_1 + ER_4$

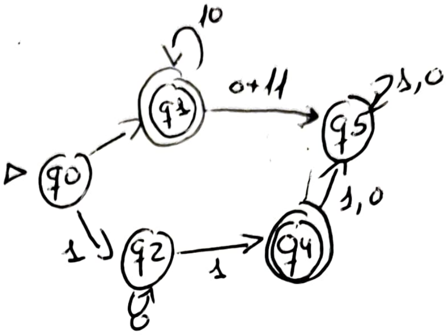
$q_0 \equiv P$ $q_2 \equiv R$ $q_4 \equiv T$
$q_1 \equiv Q$ $q_3 \equiv S$ $q_5 \equiv U$

Eliminamos q_3

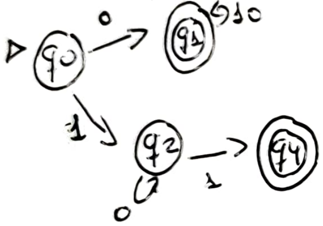
$\to q_3$	$q_3 \to$
q_1	q_1 q_5

$\overline{q_1 q_1} \equiv 10$

$\overline{q_1 q_5} \equiv 11$



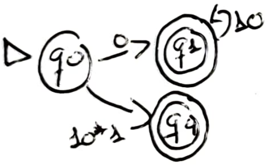
Como q_5 es un estado sumidero se puede eliminar.



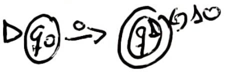
Eliminar q_2

$\to q_2$	$q_2 \to$
q_0	q_4

$\overline{q_0 q_4} \equiv 10^*1$

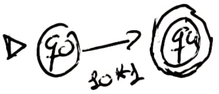


1° ER_1 eliminando q_4



$ER_1 \equiv 0(10)^*$

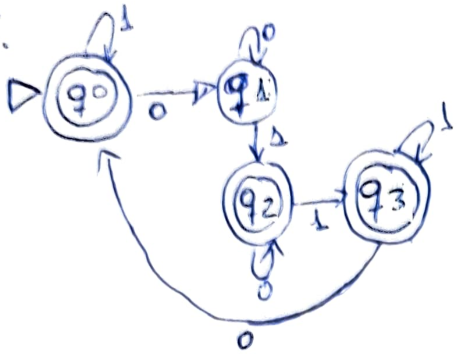
2° ER_2 eliminando q_1



$ER_2 \equiv 10^*1$

obtenemos que $ER \equiv ER_1 + ER_4 \equiv \underline{\underline{0(10)^* + 10^*1}}$

6.

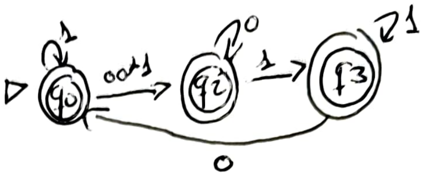


$q_0 \equiv A$ $q_1 \equiv B$, $q_2 \equiv C$, $q_3 \equiv D$

Eliminar q_1:

$\to q_1$	$q_1 \to$
q_0	q_2

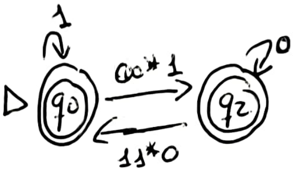
$\overline{q_0 q_2} \equiv 00^*1$



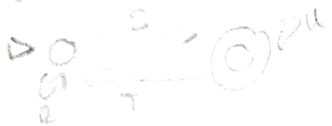
Si Elim q_3

$\to q_3$	$q_3 \to$
q_2	q_0

$\overline{q_2 q_0} \equiv 11^*0$



Ej general para reducir trabajo:



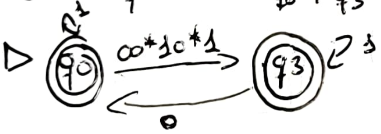
$ER = (R+SU^*T)^* SU^*$

Obtenemos que $ER_2 = (1+00^*10^*11^*0)^* 00^*10^*$

Si elim q_2

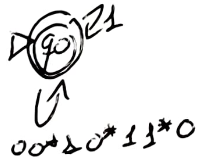
$\to q_2$	$q_2 \to$
q_0	q_3

$\overline{q_0 q_3} \equiv 00^*10^*1$



$ER_3 = (1+00^*10^*1\,1^*0)^* 00^*10^*11^*$

Si volvemos a eliminar q_3 y luego q_1:



$ER_0 = (1+00^*10^*11^*0)^*$

$ER \equiv ER_0 + ER_2 + ER_3 = (1+00^*10^*11^*0)^* + (1+00^*10^*11^*0)^* 00^*10^* + (1+00^*10^*11^*0)^*$
$00^*10^*11^* = (1+00^*10^*11^*0)^* (\varepsilon + 00^*10^* + 00^*10^*11^*) =$
$= \underline{\underline{(1+00^*10^*11^*0)^* (\varepsilon + 00^*10^*(\varepsilon + 11^*))}}$